

Experiment-19

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Course Code: -MLA0305

Slot:B

Title: Reinforcement Learning in Partially Observable Markov Decision Processes

Aim:

To implement a Reinforcement Learning agent for solving POMDP environments and evaluate decision-making under uncertainty.

Procedure:

1. Define a partially observable environment.
2. Define the hidden states and available observations.
3. Allow the agent to receive incomplete observations.
4. Maintain a history or belief representation of previous observations.
5. Select actions based on the available information.
6. Execute actions and receive rewards.
7. Update the agent's policy using the observed experience.
8. Repeat the process for multiple episodes.
9. Measure cumulative reward and decision accuracy.
10. Analyze the effect of partial observability.



The screenshot shows a Jupyter Notebook interface. At the top, there are tabs for 'Commands', 'Code', 'Text', and 'Run all'. On the right, there are indicators for 'RAM' and 'Disk' usage. The left sidebar contains icons for file management and search. The main area displays a code cell with the following Python code:

```
belief={'Good':.5,'Bad':.5}; like={'Good':{'Safe':.9,'Danger':.1},'Bad':{'Safe':.2,'Danger':.8}}; obs='Danger'
post={s:belief[s]*like[s][obs] for s in belief}; z=sum(post.values()); post={s:v/z for s,v in post.items()}
print('Prior:',belief);print('Observation:',obs);print('Posterior:',post)
```

The output of the code cell is as follows:

```
... Prior: {'Good': 0.5, 'Bad': 0.5}
Observation: Danger
Posterior: {'Good': 0.11111111111111112, 'Bad': 0.8888888888888889}
```

Result:

The RL agent successfully learned to make decisions using incomplete observations. The experiment demonstrated how Reinforcement Learning can handle uncertainty in POMDP environments.